

Final Assessment Test(FAT) - NOV/DEC 2025

Programme	B.Tech.	Semester	Fall Semester 2025-26
Course Code	BECE202L	Faculty Name	Prof. Usha Rani S
Course Title	Signals and Systems	Slot	A1+TA1
		Class Nbr	CH2025260100334
Time	3 hours	Max. Marks	100

Instructions To Candidates

- Write only your registration number in the designated box on the question paper. Writing anything elsewhere on the question paper will be considered a violation.

Course Outcomes

- CO1: Differentiate between various types of signals and understand the implication of operations on signals.
 CO2: Understand the terms like causal, dynamic, linear, time invariant and stability of systems. Also, students will be able to compute impulse response of both continuous time and discrete time systems.
 CO3: Perform the transformation of CT and DT signals from time domain to frequency domain and understand the concept of distribution of energy as a function of frequency.
 CO4: Convert the CT signals to DT signals and vice versa and understand their consequences.
 CO5: Processing of band pass signals through band pass systems.
 CO6: Solve differential and difference equations, with initial conditions, using Laplace and Z transforms respectively.

Section - I

Answer all Questions (7 × 10 Marks)

- The systems given below have input $x[n]$ and output $y[n]$. Determine whether the system is (a) Static, (b) Causal, (c) Time Invariant, (d) Linear and (e) Stable.
 $y[n] = \sin(2\pi x[n]) + x[n+1]$ (Each 2 Marks) [10] (CO2/K3)
- a) The impulse response of discrete time LTI system is given by
 $h(n) = (0.5)^n u(n)$
 If the input signal is $x[n] = \begin{cases} n+1, & 0 \leq n \leq 2 \\ 0, & \text{otherwise} \end{cases}$
 Determine the output $y(n) = x(n) * h(n)$ (4marks)
 b) Let $x(t) = e^{-t}u(t)$, $y(t) = e^{-2t}u(t)$. Find the cross correlation $R_{xy}(\tau)$ (4Marks)
 c) State and explain one practical application of correlation in signal processing. (2 Marks) [10] (CO2/K3)
- A real-valued, continuous-time periodic signal $x(t)$ has a fundamental period T_0 and its CTFS representation is given by

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

 It is known that $x(t)$ has the following symmetry: $x(t) = -x(t+T_0/2)$. This is known as half-wave symmetry.
 a) Prove that for a half-wave symmetric signal, the CTFS coefficients a_k are zero for all even values of k (i.e., $a_k = 0$ if k is even). (7 marks)
 b) The signal $x(t)$ is also real and odd. What additional simplifications can be stated about the coefficients a_k ? (3 marks) [10] (CO3/K3)
- a) "The Hilbert transform is the key to modern communication systems". Justify this statement with required expressions and show clearly how this avoids complexity in communication receivers. Also, what is the frequency response $H_h(f)$ of an ideal Hilbert transformer. [5 Marks]
 b) A signal is given by $x(t) = 2 \cos(100\pi t)$. Find its Hilbert transform $\hat{x}(t)$. [3 Marks]
 c) What would be the Hilbert transform of $\hat{x}(t)$ from part (b)? What is the significance of this result? [2 Marks] [10] (CO5/K2)

05. A signal $x_1(t)$ is band limited to 5KHz while the another signal $x_2(t)$ is band limited to 8KHz. Find the nyquist sampling rate for i) $x_1(t)x_2(t)$ ii) $x_1(t) * x_2(t)$. (Each 5 Marks)

[10] (CO4/K3)

06. Let the system define by its differential equation as $\frac{d^2}{dt^2}y(t) + \frac{d}{dt}y(t) + 2y(t) = \frac{d}{dt}x(t) + x(t)$. Assume zero initial conditions.

a) Determine the system transfer function $H(s)$. (5 Marks)

b) Assume the input $x(t) = e^{-3t}u(t)$. Find the system output $y(t)$. (5 Marks)

[10] (CO6/K3)

07. Consider a causal discrete time system whose output $y[n]$ and input $x[n]$ are related by

$$y[n] - \frac{5}{6}y[n-1] + \frac{1}{6}y[n-2] = x[n]$$

a) Find its system function $H(z)$ (5 marks)

b) Find its impulse response $h[n]$ (5 Marks)

[10] (CO6/K3)

Section - II

Answer all Questions (2 × 15 Marks)

08. a) Draw the waveform of the given signal and also sketch its derivative. (6 marks)

$$x(t) = u(t) - 2u(t-1) + u(t-2)$$

b) For the square wave signal $g(t)$ shown in Fig 1 find the component in $g(t)$ of the form $\sin t$. In other words, approximate $g(t)$ in terms of $\sin t$. So that the energy of the error signal is minimum (6 marks)

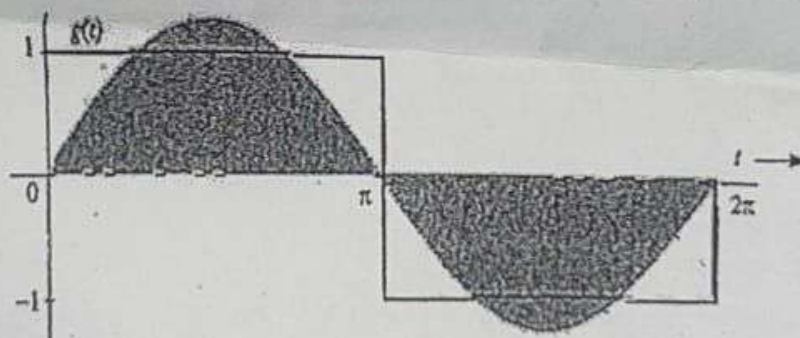


Fig.1

- c) Determine whether the following signal is energy or power signal (3marks)

$$x(t) = \text{rect}\left(\frac{t}{\tau}\right)$$

[15] (CO1/K2)

09. a) Evaluate the integral $\int_{-\infty}^{\infty} e^{-2|t|} \cos(20t) dt$ using CTFT (7 Marks)

- b) Find the inverse Discrete Time Fourier Transform to obtain $h[n]$ if (8 Marks)

$$H(e^{j\omega}) = 3 + 2 \cos(\omega) + 4 \cos(2\omega) \cos\left(\frac{\omega}{2}\right) e^{-j\frac{\omega}{2}}$$

[15] (CO3/K2)